



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 6

Equity Portfolio Management (Part 2)



STUDY NOTES • EDITION 1.0 • WEEK 6

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How to use these notes



These notes cover the whole of Week 6: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A estimates beta properly and shows what it is ultimately for — valuing an asset. Part B asks whether a portfolio actually performed, using three measures that can disagree. Part C scales the two-asset arithmetic of Week 5 up to a real multi-asset portfolio and ends on the single decision that matters most. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

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2	Reading beta and the security market line
3	Valuation from first principles

B Judging performance

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C Building a real portfolio

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9	Asset allocation

CONSOLIDATION

Module summary, by learning objective
Glossary and self-check

Module overview



Week 5 built the tools on one and two assets. This module generalises to many, and adds the two questions that follow construction: what is this asset worth, and did the portfolio actually perform? It begins by estimating beta through regression and using it to value an asset, turns to the three risk-adjusted measures of performance, and closes with the correlation matrix, multi-asset construction and asset allocation.

One thread runs through all nine topics: **a number is only useful once it is risk-adjusted**. A raw return, a raw beta and a raw allocation all mislead until they are set against the risk that produced them.

Learning objectives

By the end of this module, you will be able to:

- 1 **Estimate** beta by regression, interpret it, and locate a stock on the security market line.
- 2 **Explain** what determines an asset's value and identify the two inputs a valuation requires.
- 3 **Compute** portfolio beta and apply Jensen's alpha, the Sharpe ratio and the Treynor ratio.
- 4 **Build** a multi-asset portfolio from a correlation matrix and justify an asset allocation.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	Estimating beta by regression	A • Estimating	L20 V1	LO 1
2	Reading beta and the security market line	A • Estimating	L20 V1	LO 1
3	Valuation from first principles	A • Valuing	L21 V1	LO 2
4	Portfolio return and portfolio beta	B • Measures	L22 V1	LO 3
5	Jensen's alpha	B • Measures	L22 V2	LO 3
6	Sharpe and Treynor	B • Measures	L22 V2	LO 3
7	The correlation matrix	C • Construction	L23 V1	LO 4
8	Building a multi-asset portfolio	C • Construction	L23 V2	LO 4
9	Asset allocation	C • Construction	L23 V3	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Regression

A method measuring, on average, how one variable moves when another moves.

Residual

The vertical distance from a data point to the regression line; unsystematic risk.

R-squared

The proportion of the target's movement explained by the model.

Underpriced

A stock plotting above the security market line — more return than its risk justifies.

Cash-generating ability

An asset's second defining characteristic, alongside liquidity.

Discount rate

The rate used to convert future cash flows to present value; supplied by CAPM.

Diversification benefit

The gap between the straight line and the curved frontier.

Sharpe ratio

Excess return divided by total risk (sigma).

Correlation matrix

A square, symmetric grid holding every pair's correlation, with 1.00 on the diagonal.

Covariance

A pair's correlation multiplied by both standard deviations.

Asset allocation

The division of wealth across asset classes; the decision explaining most long-term outcome.

Alpha (intercept)

The portion of a stock's return not explained by market movement.

Least squares

The method selecting the line that minimises the sum of squared residuals.

Security market line

The plot of the CAPM equation, beta on the horizontal axis and expected return on the vertical.

Overpriced

A stock plotting below the security market line.

Present value

Today's worth of a future amount, found by discounting at rate r .

Portfolio beta

The weighted average of the constituent securities' betas.

Jensen's alpha

Actual portfolio return minus the return CAPM predicted for its beta.

Treynor ratio

Excess return divided by systematic risk (beta).

Heat map

A colour-coded correlation grid that makes sector blocks visible at a glance.

Covariance matrix

The grid of covariances used to compute portfolio variance.

Age rule of thumb

Equity exposure of 100 minus age, with modern variants using 110 or 120.

PART A • ESTIMATING AND VALUING

Where beta comes from, and what it is for

1 Estimating Beta by Regression

PART A • ESTIMATING AND VALUING • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Beta is the slope of the single straight line that best fits a cloud of paired returns — nothing more mysterious than that.

Why this matters

Week 5 used beta as a given. This topic opens it up, and the regression underneath turns out to be the ancestor of much of what Week 11 calls machine learning.

Returns, not prices

Beta tracks movement in **returns**. A price of ₹1,500 tells us little; the percentage change is what matters. So the first job is always to convert both series — the stock and the market index — to returns using $R = (P_1 - P_0) \div P_0$, computed identically and by the same method every time.

The logic of regression

Regression measures how one variable moves when another moves. Heavy rain drives umbrella sales; study hours drive marks; fertiliser drives crop yield. Map that onto beta: the Nifty 50 return is **X**, the influencer, and the stock's return is **Y**, the influenced. Beta answers one question — for every 1% change in the Nifty 50, how much does this stock move on average?

Plot sixty months as sixty dots, market return across the bottom and stock return up the side. Regression draws the single straight line passing as close as possible to all sixty at once.

The three components

Stock return = alpha + (beta × market return) + error term.

Alpha is the intercept — where the line crosses the vertical axis — and represents the stock's estimated average return in a month when the market returns nothing. *Beta* is the slope: if it is 1.30, a 1% market rise implies roughly a 1.30% rise in the stock. The *error term*, also called the residual, is the vertical distance from a point to the line — the part of that month's return the market cannot explain.

This restates a Week 5 lesson. What the line explains is **systematic risk**; what sits in the error term is **unsystematic risk**, the part diversification removes. Beta, we can now see, is simply a regression coefficient.

Least squares, and R-squared

Which line is best? Measure how far every point lies above or below a candidate line — the residuals. Square each one, so positive and negative errors cannot cancel and large errors count more heavily. Add the squared residuals across all sixty observations. Choose the line making that total as small as possible. This is the method of **least squares**, more than two hundred years old and originally developed for tracking the movement of planets.

R-squared then measures the proportion of the stock's total movement the model explains. If R^2 is 0.45, roughly 45% of the movement is explained by the broader market and the remaining 55% by company-specific factors.

ONE EQUATION, THREE PARTS

ALPHA

The intercept • return unexplained by the market

BETA × MARKET RETURN

The slope • systematic risk • what the line explains

ERROR TERM

The residual • unsystematic risk • what the line cannot explain

In Excel: =SLOPE(stock returns, market returns). Y first, X second. Swap the order and Excel returns a completely wrong beta, with no warning.

Figure 1.1 The regression equation. What the line explains is systematic; what it misses is not.

KEY IDEA

The regression line does not merely estimate beta. It *partitions* the stock's movement into the part the market caused and the part it did not — which is precisely the systematic and unsystematic split that justifies diversification in the first place.

EXAM POINTER

Name all three components with what each represents, then describe least squares in the four steps — measure, square, sum, minimise. State why squaring is necessary: it stops errors cancelling and penalises large deviations. If R^2 is given, interpret both halves, not just the explained one.

QUICK RECAP • TOPIC 1

- Beta tracks returns, never prices; convert both series first.
- Stock return = alpha + (beta × market return) + error term.
- Alpha is the intercept, beta the slope, and the error term is unsystematic risk.
- Least squares selects the line minimising the sum of squared residuals.
- R^2 of 0.45 means 45% of movement is explained by the market and 55% is not.

2 Reading Beta and the Security Market Line

PART A • ESTIMATING AND VALUING • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

A beta of 1.4 means the stock amplifies the market by about forty per cent in both directions — and where it plots against the security market line tells you whether it is fairly priced.

Why this matters

Beta interpretation is the bridge from estimation to decision. The security market line in particular is how CAPM stops being an equation and becomes a verdict on whether to buy.

What the number means

Running =SLOPE on ABC Ltd against the Nifty 50 returns a beta of **1.4**. ABC moves about 1.4% for every 1% move in the index — up about 14% when the market rises 10%, and down about 14% when it falls 10%. It amplifies the market.

Reading any beta follows the same scale. Above 1 is aggressive, more volatile than the market. Exactly 1 moves with it. Between 0 and 1 is defensive, muting the swings. Below 0 is rare and prized, moving opposite to the market and acting as natural insurance.

The security market line

The security market line plots the CAPM equation, with beta on the horizontal axis measuring systematic risk and expected return on the vertical. As beta increases, expected return rises, because investors demand greater reward for greater risk.

The **intercept is R_f** , the risk-free rate, where beta is zero. The **slope is the market risk premium**, calculated as the market return minus R_f , and its steepness reflects how much extra return each unit of beta earns.

Position tells you pricing

A stock sitting **on** the line is fairly priced, offering exactly the return its risk justifies. **Above** the line it is underpriced — delivering more return than its beta warrants, and therefore attractive. **Below** the line it is overpriced.

This is how the model becomes a decision rule. Estimate beta, compute the CAPM required return, compare it with what the stock actually offers, and the gap tells you which side of the line it sits on.

WHERE A STOCK PLOTS AGAINST THE LINE IS A VERDICT ON ITS PRICE

ABOVE THE LINE • UNDERPRICED

Delivers more return than its beta justifies • attractive

ON THE LINE • FAIRLY PRICED

Exactly the return its systematic risk warrants

BELOW THE LINE • OVERPRICED

Delivers less return than its beta justifies

*Intercept = R_f . Slope = **the market risk premium**. The line itself is just CAPM, plotted.*

Figure 2.1 The security market line. Position relative to it is the pricing verdict.

COMMON PITFALL

Confusing the security market line with the efficient frontier. The frontier plots return against *total* risk (sigma) and curves; the security market line plots return against *systematic* risk (beta) and is straight. They answer different questions.

EXAM POINTER

Given a beta and an actual expected return, always compute the CAPM required return first, then compare. State the intercept and slope of the line explicitly. A question giving a stock above the line wants the word *underpriced* and the reason — more return than its systematic risk justifies.

QUICK RECAP • TOPIC 2

- A beta of 1.4 means the stock moves about 1.4% for every 1% market move.
- Above 1 is aggressive, 1 tracks the market, 0 to 1 is defensive, below 0 is insurance.
- The security market line plots expected return against beta.
- Its intercept is the risk-free rate; its slope is the market risk premium.
- On the line is fairly priced, above it underpriced, below it overpriced.

3 Valuation from First Principles

PART A • ESTIMATING AND VALUING • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

An asset is worth the present value of all the future cash flows it is expected to generate — and CAPM supplies the rate at which you discount them.

Why this matters

This is where beta earns its keep. The whole apparatus of Weeks 5 and 6 exists to produce one number, r , and this topic shows what that number is for.

Why anyone buys an asset

A plot of land costs ₹50 lakh. Gold trades at ₹10,000 a gramme. A share is quoted at ₹1,200. Where do these numbers come from? Go back further: why would anyone buy an asset at all?

Because we expect it to create economic value in the future. This is the second defining characteristic of an asset — we met liquidity earlier, and now we add **cash-generating ability**. Education costs money now and pays back as higher lifetime earnings. A laptop is not a gadget but a tool that helps generate income. An Infosys share is a claim on a company's future cash flows. A city flat produces rent or resale.

The value of an asset is the present value of all the future cash flows it is expected to generate. Price is what we observe; valuation is our attempt to explain it.

Why we need a shared model

Give several artists the same scene, the same brushes and the same paints, and every canvas comes out different. Hand the same company to two skilled analysts and you get two numbers, every time.

A shared model does not force everyone to an identical answer. It ensures everyone follows the same consistent approach, making results comparable and trustworthy. We chase a method that is reliable, repeatable and shared — not perfection.

The two inputs, and where the second comes from

Valuation needs exactly two things. The **cash flow** is given to you in this course. The **discount rate, r** , is the only input left to find — and it comes from CAPM.

That is the crucial link. The beta estimated for ABC Ltd against the Nifty 50 in Topic 1 feeds into CAPM, which produces r , which discounts the future cash flows. Last week's tool becomes this week's input.

$PV = CF \div (1 + r)^n$. Each cash flow is discounted by how far away it sits in time. A single-period asset has one flow discounted once; over many periods, sum every flow — and the further out a flow sits, the heavier its discount.

THE CHAIN THAT CONNECTS THE WHOLE MODULE



Higher discount rate, lower present value. At $r = 0$ there is no discounting and ₹1,000 a year from now is worth a full ₹1,000. At 30% over one year it is worth ₹769; held seven years, about ₹159.

Figure 3.1 From price data to a valuation. Every step exists to produce r .

KEY IDEA

The discount rate is powerful out of all proportion to its size. Small changes in r swing today's value dramatically — which is exactly why the CAPM inputs that produce it deserve the care Weeks 5 and 6 give them.

COMMON PITFALL

Treating valuation and price as the same thing. Price is observed in the market; valuation is a model's estimate of what the price ought to be. The gap between them is the entire basis of active investing.

EXAM POINTER

State the definition of value in full — the present value of expected future cash flows — then write $PV = CF \div (1 + r)^n$ and name both inputs. The mark most often missed is saying explicitly that CAPM supplies r ; many answers leave the discount rate unexplained.

QUICK RECAP • TOPIC 3

- An asset is worth the present value of the future cash flows it is expected to generate.
- Cash-generating ability is an asset's second defining characteristic, alongside liquidity.
- A shared model gives comparability, not identical answers.
- $PV = CF \div (1 + r)^n$; value and discount rate move inversely.
- CAPM supplies r , which is why beta estimation matters for valuation.

PART B · JUDGING PERFORMANCE

Did the portfolio actually perform?

4 Portfolio Return and Portfolio Beta

PART B · JUDGING PERFORMANCE · TOPIC 4 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

Portfolio return and portfolio beta are both clean weighted averages. Portfolio risk is not — and that single asymmetry explains the curve.

Why this matters

This topic resolves something Week 5 left implicit: why sigma refuses to add up while beta does. The answer shapes every risk-adjusted measure that follows.

Return behaves simply

$R_p = W_1 R_1 + W_2 R_2 + \dots + W_n R_n$. Clean, intuitive, and it always adds up.

Risk refuses to play by the same rule, because correlation gets in the way. When two stocks do not move in step, some ups and downs cancel — Reliance and Infosys rarely rise and fall in unison, so holding both smooths the ride.

Why the frontier curves

If risk added up like return, the plot of risk against return would be a straight line. It bends because imperfectly correlated assets offset each other. That curvature is the visible fingerprint of diversification, and the shaded gap between the straight line and the curve is the **diversification benefit** — you get the combined return without fully inheriting the combined risk.

Why beta does add up

CAPM assumes investors hold well-diversified portfolios, so unsystematic risk has been diversified away to almost nothing. What remains is systematic risk, which by definition is the shared market factor — and shared factors combine linearly.

$B_p = \sum (W_i \times \beta_i)$. Multiply each weight by its security's beta and add. Once company-specific noise is stripped away, a tangled calculation collapses into a simple sum.

PORTFOLIO BETA • A WEIGHTED AVERAGE THAT ACTUALLY WORKS

Security	Weight	Beta	Contribution
Security 1	0.50	1.20	$0.50 \times 1.20 = 0.60$
Security 2	0.30	0.80	$0.30 \times 0.80 = 0.24$
Security 3	0.20	1.50	$0.20 \times 1.50 = 0.30$
Portfolio beta	1.00	—	$0.60 + 0.24 + 0.30 = 1.14$

Compare with sigma, which needs a correlation term for every pair. **Beta adds; sigma does not.** That is the whole difference between the two measures.

Figure 4.1 Portfolio beta. Three multiplications and a sum, because correlation has already been diversified away.

KEY IDEA

Sigma and beta behave differently under aggregation for one reason: sigma still contains the company-specific noise that interacts between assets, and beta does not. Remove the noise and the arithmetic becomes linear.

EXAM POINTER

Compute portfolio beta as a weighted sum and show every product before adding. Then state, in one sentence, why the same approach fails for sigma — correlation interferes. That contrast is what the question is really testing.

QUICK RECAP • TOPIC 4

- $R_p = \sum(W_i R_i)$ — portfolio return is a simple weighted average.
- Portfolio sigma is not, because correlation interferes between every pair.
- That interference is what bends the frontier and creates the diversification benefit.
- $B_p = \sum(W_i \times \beta_i)$ — portfolio beta is also a clean weighted average.
- Weights of 0.5, 0.3, 0.2 on betas 1.2, 0.8, 1.5 give a portfolio beta of 1.14.

5 Jensen's Alpha

PART B • JUDGING PERFORMANCE • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

CAPM gives every portfolio a personalised benchmark. Alpha is how far above or below it the manager actually landed.

Why this matters

Two athletes run ten kilometres. Arjun finishes in 48 minutes, Rohan in 52. Arjun looks better — until you learn Arjun ran on a flat road and Rohan ran uphill carrying a twenty-kilogramme backpack.

Why a raw return proves nothing

Two fund managers may both report 18%. One earned it through genuine skill; the other took enormous risks and got lucky. A return number on its own can never tell you which is which.

CAPM solves this by telling us the return a portfolio *should* have earned for the systematic risk it carried. That makes it a personalised benchmark: the risk-free rate and the market return are the same for everyone, so only beta varies, and each portfolio gets a benchmark set purely by its own risk.

The formula

$\text{Alpha} = R_p - [R_f + \beta_p \times (R_m - R_f)]$ — the actual return minus the return CAPM predicted.

Alpha = 0 means no value added; the portfolio earned exactly what its risk predicted. **Alpha > 0** means genuine skill — the manager beat the benchmark after adjusting for risk. **Alpha < 0** means value destroyed, however impressive the headline return.

The honest story

Take ABC Growth Fund: $R_f = 6\%$, $\beta = 1.2$, $R_m = 13\%$, actual return 18%. The CAPM predicted return is $6\% + 1.2 \times (13\% - 6\%) = 6\% + 8.4\% = 14.4\%$. Alpha is $18\% - 14.4\% = +3.6\%$.

Notice what that does to the story. The raw gap versus the market is 5 percentage points — 18% against 13%. The risk-adjusted excess is only 3.6%. Alpha strips out the portion that was simply market exposure amplified by a beta above one.

SPLITTING A HEADLINE RETURN INTO EXPOSURE AND SKILL

Step	Working	Result
Market risk premium	$13\% - 6\%$	7%
Security risk premium	$1.2 \times 7\%$	8.4%
CAPM predicted return	$6\% + 8.4\%$	14.4%
Actual return	Reported by the fund	18%
Jensen's alpha	$18\% - 14.4\%$	+3.6%

*The fund beat the market by **5 points**. It beat its own risk-adjusted benchmark by only **3.6**. The difference is not skill — it is beta.*

Figure 5.1 Jensen's alpha. The raw outperformance and the risk-adjusted one are different numbers.

KEY IDEA

A high beta manufactures outperformance in a rising market without any skill at all. Alpha exists to strip that out — which is why a fund advertising its raw return in a bull market is advertising its beta.

EXAM POINTER

Compute the CAPM benchmark as a separate step before subtracting; examiners award it separately. State the sign interpretation for all three cases. The strongest answers note that the raw market-beating gap and the alpha are different figures, and explain why.

QUICK RECAP • TOPIC 5

- A raw return cannot distinguish skill from risk-taking.
- CAPM supplies a personalised benchmark, since only beta varies between portfolios.
- Alpha = actual return – CAPM predicted return.
- Positive alpha means skill, zero means none added, negative means value destroyed.
- R_f 6%, β 1.2, R_m 13%, actual 18% gives a CAPM return of 14.4% and an alpha of +3.6%.

6 Sharpe and Treynor

PART B • JUDGING PERFORMANCE • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Two ratios, one numerator, different denominators — and which one to use depends entirely on whether the portfolio is all the investor owns.

Why this matters

Alpha answers whether a manager beat a benchmark. These two answer a different question: how much excess return was earned per unit of risk taken. The rankings can disagree, which is the point.

The Sharpe ratio

Sharpe = $(R_p - R_f) \div \sigma_p$ — excess return per unit of **total** risk. The denominator is the portfolio's full standard deviation, not beta. Higher is always better.

With a risk-free rate of 6%: ABC Growth Fund returns 18% at a sigma of 20%, giving $(18 - 6) \div 20 = \mathbf{0.60}$. XYZ Balanced Fund returns 16% at a sigma of 12%, giving $(16 - 6) \div 12 = \mathbf{0.83}$.

XYZ wins despite the lower headline return, because it delivered more return per unit of risk. A higher return is not automatically better — once risk is accounted for, a rational investor prefers the fund that earned less.

The Treynor ratio

Treynor = $(R_p - R_f) \div \beta_p$ — the same excess return on top, divided by systematic risk only. For ABC: $(18 - 6) \div 1.2 = \mathbf{10.0}$. The only change from Sharpe is the denominator.

Total risk splits into two parts. Unsystematic risk is diversifiable — a strike, a lawsuit, a bad product — and shrinks as stocks are added, so the risk curve falls towards a floor. Systematic risk is that floor: recessions, rate moves, wars, forces hitting every stock at once. The curve flattens but never reaches zero, and beta captures the floor.

Which one to use

Use **Sharpe** when the portfolio is the investor's *entire wealth*. Total risk is what truly matters, because there is nothing else to cushion a bad outcome.

Use **Treynor** when the portfolio is *one part of a larger diversified holding*. Company-specific risk has already been diversified away elsewhere, so only systematic risk is relevant to judging this slice.

THREE MEASURES ON THE SAME FUND • THEY CAN DISAGREE, AND THAT IS INFORMATIVE

Measure	Formula	ABC result	How to read it
Jensen's alpha	$R_p - \text{CAPM predicted return}$	+3.6%	Earned 3.6% above the risk-adjusted benchmark — the manager added value
Sharpe ratio	$(R_p - R_f) \div \sigma_p$	0.60	0.60% of excess return per 1% of total risk
Treynor ratio	$(R_p - R_f) \div \beta_p$	10.0	10 units of excess return per unit of systematic risk

Comparing one fund across all three gives the complete picture, because **the ranking can shift** depending on how risk is defined.

Figure 6.1 The three risk-adjusted measures. Same fund, three different questions.

REAL-WORLD EXAMPLE

ABC returns 18% and XYZ returns 16%. On raw return ABC wins. On Sharpe, XYZ wins at 0.83 against 0.60. Both statements are true, and an investor holding only one of these funds should prefer XYZ — the extra two points of return came with eight extra points of volatility.

EXAM POINTER

Compute both ratios and state the denominator difference explicitly — total risk against systematic risk. The high-value mark is the selection rule: Sharpe when the portfolio is everything the investor owns, Treynor when it is one component of a diversified holding.

QUICK RECAP • TOPIC 6

- Sharpe = $(R_p - R_f) \div \sigma_p$ — excess return per unit of total risk.
- Treynor = $(R_p - R_f) \div \beta_p$ — excess return per unit of systematic risk.
- ABC scores 0.60 on Sharpe against XYZ's 0.83, despite ABC's higher raw return.
- Use Sharpe when the portfolio is the investor's entire wealth.
- Use Treynor when it is one slice of a larger diversified holding.

PART C • BUILDING A REAL PORTFOLIO

From two assets to a real portfolio

7 The Correlation Matrix

PART C • BUILDING A REAL PORTFOLIO • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

With two stocks one number told the whole story. With a hundred there are 4,950 pairs — which is why the matrix exists.

Why this matters

Part C moves from two assets to many. The matrix is the raw material for everything that follows, and the counting formula explains why it cannot be done by hand.

Why pairs explode

With two stocks a single CORREL result sufficed. With four, every stock relates to every other, and one relationship becomes many. **Unique pairs** = $n \times (n - 1) \div 2$.

Four stocks give six pairs — manageable by hand. Ten give forty-five — already a crowded list. Fifty give **1,225**. A hundred give **4,950**. Pairs grow far faster than stocks, and no analyst can absorb that by reading individual values.

Three rules for reading it

The matrix is a square table where each row and column is a stock, and every cell holds the correlation between that row stock and that column stock.

The diagonal is always 1.00, because every stock is perfectly correlated with itself — it is the spine of the grid. **The grid is symmetric**: A with B equals B with A, mirrored across the diagonal. **Lower is better**: the closer to -1, the greater the risk reduction.

A 4×4 grid therefore has sixteen cells but needs only six CORREL formulas — four diagonal cells typed in as 1.0, six unique pairs computed, and six mirror cells copied.

The heat map

Investment teams read matrices as **heat maps**: colour-coded grids where low values are green and high values red. Patterns emerge instantly — four banks may form a red block moving together at around 0.75, three IT companies another. The green areas between the blocks are what make diversification useful. One heat map replaces dozens of separate calculations.

FOUR STOCKS, SIX UNIQUE PAIRS

Pair	Correlation	Reading
ABC – XYZ	0.33	Moderate positive — they drift the same way, but weakly. The pair from Week 5
ABC – PQR	0.65	High positive — when ABC rises, PQR tends to rise strongly alongside it
XYZ – LMN	0.71	Strongly linked together
ABC – LMN	-0.12	Negative — the crown jewel. LMN tends to rise when the rest is bleeding

*Few real assets are negatively correlated, which is exactly why they are prized. Among these four, **LMN is the best single diversifier** — think of gold, which often moves inversely to Indian equities.*

Figure 7.1 Selected pairs from a four-stock matrix. Lower is better, and negative is rare.

COMMON PITFALL

Computing all sixteen cells of a 4×4 matrix. Four are always 1.00 and six are mirrors of the other six. Doing the redundant work is not wrong, but in a fifty-stock matrix it is the difference between 1,225 formulas and 2,500.

EXAM POINTER

State the counting formula and apply it — questions frequently ask for the number of pairs at a given n . Name all three reading rules, and say why a negative correlation is prized rather than merely noting that it exists.

QUICK RECAP • TOPIC 7

- Unique pairs = $n \times (n - 1) \div 2$. Four stocks give six; a hundred give 4,950.
- The diagonal is always 1.00 and the grid is symmetric across it.
- Lower correlation means greater risk reduction; negative is best and rare.
- A 4×4 grid needs only six CORREL formulas, not sixteen.
- Heat maps colour-code the grid so sector blocks become visible instantly.

8 Building a Multi-Asset Portfolio

PART C • BUILDING A REAL PORTFOLIO • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

Four stocks, four weights, and one matrix multiplication — and the resulting risk sits well below the naive average.

Why this matters

This is the computational core of the module and the point at which portfolio theory stops being a two-asset illustration and becomes something a desk could actually run.

The inputs

A portfolio works like a biryani: four ingredients, each with its own character, and the dish that emerges is decided by the proportions chosen.

Stock	Weight	Expected annual return	Standard deviation
ABC Ltd	30% (0.30)	14%	18%
XYZ Ltd	25% (0.25)	11%	14%
PQR Ltd	25% (0.25)	16%	22%
LMN Ltd	20% (0.20)	9%	10%
Total	100% — fully invested	—	—

Return is easy; risk is not

Portfolio return is a weighted average, computed in Excel with a single **=SUMPRODUCT(weights, returns)**, giving 0.1275 — **12.75%**.

The tempting mistake is to average the four sigmas the same way: $(0.30 \times 18) + (0.25 \times 14) + (0.25 \times 22) + (0.20 \times 10) = 16.4\%$. That is wrong, because the four stocks do not all move together. When one dips another may rise, and those offsetting moves cancel some of the swings.

From correlation to covariance, and the matrix route

$\text{Cov}(i,j) = \rho(i,j) \times \sigma_i \times \sigma_j$, with variance (σ^2) on the diagonal. Because ABC and LMN have a negative correlation, their covariance is negative too.

Then in Excel: build the covariance matrix; enter the weights as a column; compute **=MMULT(TRANSPOSE(weights), covariance matrix)** confirmed with Ctrl+Shift+Enter; multiply that result by the weights again with a second MMULT to get the portfolio variance; and take **=SQRT(variance)** to convert it into a risk percentage.

THE RESULT • DIVERSIFICATION MEASURED, NOT ASSERTED

Quantity	Value	What it tells you
Portfolio return	12.75%	The weighted average of the four stock returns
Naive weighted sigma	16.4%	What you would get if risk averaged — it does not
Portfolio variance	0.0146	An intermediate figure, not interpreted directly
Portfolio standard deviation	$\sqrt{0.0146} = 12.08\%$	The real risk, accounting for all correlations
Nifty 50 for reference	About 15–18%	The portfolio sits below a typical index

12.08% against a naive 16.4% and an index at 15–18%. **Diversification is genuinely removing risk**, and the matrix is how you prove it rather than assert it.

Figure 8.1 The four-asset result. The gap between 16.4% and 12.08% is the diversification benefit.

KEY IDEA

The covariance matrix is the object that makes portfolio theory scale. Two assets need one correlation; a hundred need 4,950 — but the matrix multiplication is the same two operations regardless of size.

EXAM POINTER

Compute the portfolio return by hand and show the products. For risk, state clearly that the naive weighted average is wrong and why, give the covariance formula, and name the Excel route: TRANSPOSE, MMULT twice, then SQRT. Remember that variance must be square-rooted — answers stopping at 0.0146 lose the final mark.

QUICK RECAP • TOPIC 8

- Portfolio return uses =SUMPRODUCT and gives 12.75% for the four-stock example.
- Averaging the four sigmas gives 16.4%, which is wrong because assets do not move together.
- $\text{Cov}(i,j) = \rho(i,j) \times \sigma_i \times \sigma_j$, with variance on the diagonal.
- Portfolio variance uses TRANSPOSE and two MMULT operations; SQRT converts it to risk.
- The result is 12.08% — below both the naive average and a typical index.

9 Asset Allocation

PART C • BUILDING A REAL PORTFOLIO • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

More than ninety per cent of the variation in a portfolio's long-term returns is explained by asset allocation — not by which stocks were picked.

Why this matters

The closing topic of the module, and the one with the largest practical consequence. Everything before it refines the tools; this decides where they are pointed.

The landmark finding

Brinson, Hood and Beebower (1986) found that more than 90% of the variation in a portfolio's long-term returns is explained by asset allocation alone — not stock picking, not market timing, not frantic trading. The hours spent hunting for the next Infosys matter far less than the calm, upfront decision about how money is split across asset classes.

The four classes

Class	Return potential	Risk	Defining feature
Equity	12–18%	High	The strongest long-horizon wealth creator; volatile short term
Debt	6–10%	Low to medium	Stable and predictable, but limited upside
Gold	8–12%	Medium	Rises precisely when equities fall; deep cultural significance in India
Real estate	8–15%	Medium to high	Rental income plus appreciation, but relatively illiquid

No asset class wins in every season. Equity leads in a bull market; debt and gold outperform in a recession; real assets hold value under high inflation; government bonds are the safe haven in a crisis. You cannot predict the season, so you combine the classes.

The decision framework, and the rule of thumb

Four questions shape any allocation. *What is my goal?* The goal sets the return needed, which drives the risk that must be taken. *What is my horizon?* Longer horizons tolerate more volatility. *What is my risk tolerance?* Even if you can afford risk, can you emotionally handle a 30% crash? *What are my constraints?* Liquidity, tax and legal preferences all shape the mix.

The classic shortcut is **equity % = 100 – your age**: 75% at twenty-five, 55% at forty-five, 35% at sixty-five. Treat it as a rough guideline. With longer lives and higher inflation, planners now often use 110 or even 120 minus age. The principle holds either way — the older you are, the less time you have to recover from a downturn.

THREE INVESTORS • THE SAME FOUR CLASSES, RESHAPED BY HORIZON

	Young professional, 25	Middle-aged, 45	Retired, 65
Risk and horizon	High risk, long horizon	Moderate risk, medium horizon	Low risk, short horizon
Equity	70%	50%	20%
Debt	15%	25%	55%
Gold	10%	15%	15%
Real estate	—	10%	10%
Rationale	Volatility is manageable over decades; heavy equity maximises wealth creation	Balancing growth with protection; equity still dominant, debt adds stability	Capital preservation first; heavy debt for income, some equity for inflation

The same four classes appear for all three. Age, horizon and risk tolerance reshape the entire mix — proportion is everything.

Figure 9.1 Three allocations. Note that no investor holds only one class.

KEY IDEA

Allocation is simply choosing your point on the efficient frontier. Everything in Weeks 5 and 6 — return and risk, correlation, the frontier, beta — exists to make that one choice an informed one rather than a guess.

COMMON PITFALL

Quoting the 90% finding as though it means allocation explains 90% of the *difference between* funds. It concerns variation in a single portfolio's returns *over time*. The distinction matters, and the original authors later stressed it.

EXAM POINTER

Name all four asset classes with risk level and one defining feature. Give the four framework questions, and state the age rule with at least one modern variant. If asked to allocate for a named investor, justify each figure by horizon and tolerance rather than simply listing percentages.

QUICK RECAP • TOPIC 9

- Brinson, Hood and Beebower (1986): over 90% of long-term return variation comes from asset allocation.
- Four classes: equity, debt, gold, real estate — none wins in every season.
- The framework asks about goal, horizon, risk tolerance and constraints.
- Equity % = 100 – age, with modern variants using 110 or 120.
- Allocation is the act of choosing your point on the efficient frontier.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Estimate beta by regression and locate a stock on the security market line	Topics 1 and 2 · Figures 1.1, 2.1
2	Explain what determines value and identify the two valuation inputs	Topic 3 · Figure 3.1
3	Compute portfolio beta and apply alpha, Sharpe and Treynor	Topics 4, 5 and 6 · Figures 4.1, 5.1, 6.1
4	Build a multi-asset portfolio and justify an allocation	Topics 7, 8 and 9 · Figures 7.1, 8.1, 9.1

The module in six sentences

Beta is the slope of the least-squares line fitted through a cloud of paired returns, and the residuals left over are precisely the unsystematic risk that diversification removes. Where a stock plots against the security market line — whose intercept is the risk-free rate and whose slope is the market risk premium — tells you whether it is fairly priced, and CAPM supplies the discount rate that turns expected cash flows into a valuation. Portfolio return and portfolio beta are both clean weighted averages, while portfolio sigma is not, and that asymmetry is what bends the efficient frontier. Three measures then judge performance: Jensen's alpha against a personalised CAPM benchmark, the Sharpe ratio per unit of total risk, and the Treynor ratio per unit of systematic risk. Scaling beyond two assets needs a correlation matrix, whose pairs grow as n times n minus one over two, converted to covariances and multiplied through to give a four-stock portfolio risk of 12.08% against a naive 16.4%. And more than ninety per cent of long-term return variation comes not from any of this refinement but from the initial split across asset classes.

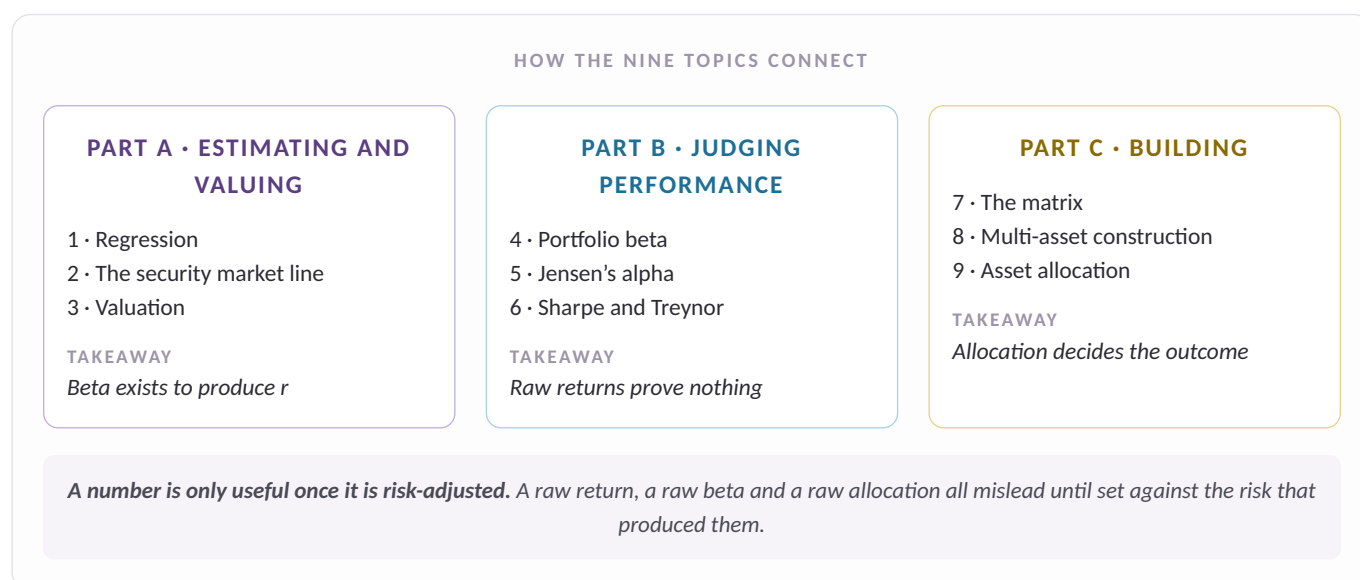


Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check



Term	Definition
Regression	A method measuring, on average, how one variable moves when another moves.
Alpha (intercept)	The portion of a stock's return not explained by market movement.
Residual	The vertical distance from a data point to the regression line; unsystematic risk.
Least squares	The method selecting the line that minimises the sum of squared residuals.
R-squared	The proportion of the target's movement explained by the model.
Security market line	The plot of the CAPM equation, beta on the horizontal axis and expected return on the vertical.
Underpriced	A stock plotting above the security market line — more return than its risk justifies.
Overpriced	A stock plotting below the security market line.
Cash-generating ability	An asset's second defining characteristic, alongside liquidity.
Present value	Today's worth of a future amount, found by discounting at rate r .
Discount rate	The rate used to convert future cash flows to present value; supplied by CAPM.
Portfolio beta	The weighted average of the constituent securities' betas.
Diversification benefit	The gap between the straight line and the curved frontier.
Jensen's alpha	Actual portfolio return minus the return CAPM predicted for its beta.
Sharpe ratio	Excess return divided by total risk (σ).
Treynor ratio	Excess return divided by systematic risk (β).
Correlation matrix	A square, symmetric grid holding every pair's correlation, with 1.00 on the diagonal.
Heat map	A colour-coded correlation grid that makes sector blocks visible at a glance.
Covariance	A pair's correlation multiplied by both standard deviations.
Covariance matrix	The grid of covariances used to compute portfolio variance.
Asset allocation	The division of wealth across asset classes; the decision explaining most long-term outcome.
Age rule of thumb	Equity exposure of 100 minus age, with modern variants using 110 or 120.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 A portfolio holds three securities at weights 0.5, 0.3 and 0.2 with betas of 1.2, 0.8 and 1.5. What is the portfolio beta, and why can you not compute portfolio sigma the same way?

1.14. $(0.5 \times 1.2) + (0.3 \times 0.8) + (0.2 \times 1.5) = 0.60 + 0.24 + 0.30$. Sigma cannot be averaged because correlation interferes between every pair; beta can, because unsystematic risk has already been diversified away.

Q2 A fund returned 18% with a beta of 1.2. The risk-free rate is 6% and the market returned 13%. Did the manager add value?

Yes — alpha is +3.6%. CAPM predicted $6\% + 1.2 \times (13\% - 6\%) = 14.4\%$. Actual 18% minus 14.4% gives 3.6%. Note the fund beat the market by 5 points but its risk-adjusted benchmark by only 3.6 — the difference is beta, not skill.

Q3 Fund A returns 18% with sigma 20%; Fund B returns 16% with sigma 12%. The risk-free rate is 6%. Which is better for an investor whose entire savings sit in one fund?

Fund B. Sharpe for A is $(18 - 6) \div 20 = 0.60$; for B it is $(16 - 6) \div 12 = 0.83$. When the portfolio is the investor's whole wealth, total risk is what matters, so Sharpe is the right measure — and the lower-returning fund wins.

Q4 How many unique correlations must you compute for a fifty-stock portfolio, and how many cells does the matrix have?

1,225 unique pairs; 2,500 cells. $n \times (n - 1) \div 2 = 50 \times 49 \div 2$. Of the 2,500 cells, fifty are the diagonal at 1.00 and the remaining 1,225 are mirrors.

Q5 Four stocks have sigmas of 18%, 14%, 22% and 10% at weights of 0.30, 0.25, 0.25 and 0.20. Why is the portfolio risk 12.08% rather than 16.4%?

Because 16.4% is the naive weighted average, and risk does not average. The covariance matrix accounts for how the four move together, and since they do not move in lockstep — ABC and LMN are actually negatively correlated — offsetting movements cancel part of the risk.

Q6 Why does asset allocation matter more than stock selection?

Because Brinson, Hood and Beebower (1986) found it explains over 90% of the variation in long-term returns. Note the precise claim: variation in a single portfolio's returns over time, not variation between funds. The practical implication is that the upfront split across classes deserves more attention than the search for individual winners.